EMBEDDED HACKING

Reverse Engineering the Raspberry Pi Pico 2

Week 4-IDA: IDA Pro — Resolve, Hack, and Patch the RP2350 (Raw `.bin`)

ARM Cortex-M33 • GDB • Ghidra

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What You'll Learn This Week

- Build the two lesson projects with `Release` and get both an `.elf` and a raw `.bin`
- Dump the **ELF symbol map** with `arm-none-eabi-nm` and use it as ground truth
- Load the raw `.bin` into IDA at `0x10000000`
- **Resolve the functions in the IDA GUI** using the ELF symbol map
- **Patch** the bytes that control the behavior, export the image, and export it

How This Guide Works

The build produces two files for each project:

File	What it is	How we use it
`.elf`	The linked image with a full symbol table	Ground truth for every function address and name
`.bin`	The raw flash image, no headers, no symbols	The image we load into IDA and reverse

The `.bin` is built **from** the `.elf`, so the ELF tells you exactly what is at every address. We use the ELF symbol map to resolve functions in IDA, and we reverse-engineer the raw `.bin` the way a real extracted firmware image is reversed.

> **Build `Release`, not `Debug`.** Every address in this guide matches the Week 4 lesson, and the Week 4 lesson is a `Release` build. `Release` optimizes the code the same way the original lesson was built: it folds `age = 42` away in Project 1 and inlines `blink_and_print` into `main` in Project 2. If you build `Debug`, the SDK function addresses move and Project 2 keeps a separate `blink_and_print`, so nothing lines up. Always build `Release` for this lesson.

The workflow is static-only: import the raw image, recover the code, decompile it, and patch the artifact:

d prove what the code does. ge. 3. Resolve the functions in IDA using the ELF symbol map. 4. Patch the bytes, export, convert and export.

Project	Prints	Also does	The hack
`0x0005_intro-to-variables`	`age: 43`	loops on `printf`	change `43` to `70`
`0x0008_uninitialized-variables`	`age: 0`	blinks the red LED on GPIO 16	change `0` to `66`, move the LED to GPIO 17

> **Addresses come from your build.** Every address here is from the `Release` build produced in Step 3. Confirm against your own `.elf` with the command in Step 4.

Part 1: Build and Get the Symbol Map

Step 1: Install the toolchain

Windows x64

- Install the **Raspberry Pi Pico** extension in VS Code. It installs the ARM GNU toolchain, CMake, Ninja, and the Pico SDK.
- Install **IDA Pro** and complete its license activation.
- Install **PuTTY** for the serial monitor.

macOS Apple Silicon

All must report `arm64`. If any is `x86_64`, put the Apple Silicon prefix first for the session and check again:

```
export PATH="/opt/homebrew/bin:$PATH"
hash -r
file "$(which cmake)"
```

Glossary

Term	Definition
`.bss`	Section for uninitialized global variables; zeroed by startup code
`.data`	Section for initialized global variables; copied from flash to SRAM at boot
`.elf`	Linked image with the symbol table; the ground truth for addresses and names
`.rodata`	Read-only section for constants and string literals; stays in flash
GPIO	General Purpose Input/Output — controllable pins on the microcontroller
ly flash	
Inlining	The optimizer replacing a function call with the function body; why `blink_and_print` disappears in `Release`

Term	Definition
Literal pool	A block of 32-bit constants that Thumb-2 code reaches with PC-relative `ldr`
MMIO	Memory-mapped I/O — hardware registers accessed as memory addresses
SIO	Single-cycle I/O — the fast GPIO block in the RP2350, at `0xd0000000`
Thumb bit	Bit 0 of a Cortex-M function pointer; selects Thumb instruction mode
UF2	USB Flashing Format — the file format the Pico 2 bootloader accepts
Vector table	The first words of flash: initial stack pointer and exception vectors

Remember: the ELF tells you what every address is, and the `.bin` is what you actually patch. Verify the static behavior from the recovered instructions.